

Sophia Shi

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Summary

- A passionate Data Engineer with 10+ years of experience building scalable cloud data pipelines and enterprise data warehouse solutions. Designed and deployed 200+ production pipelines processing over 600M rows daily, supporting 200+ business users, and reducing pipeline runtime by 40% through performance optimization and automation.
- Expert in modern cloud data platforms and ETL automation, with strong expertise in SQL, Snowflake, Azure Data Factory, and Alteryx to deliver reliable, scalable, and business-driven analytics solutions.
- Core Skills: SQL | Snowflake | Azure Data Factory | Alteryx | ETL/ELT | Data Pipelines | Azure | Logic Apps | Tableau | NetSuite | Python

Technical Skills

Database	MS SQL Server 2008 R2/2012/2017/18/20, Snowflake, NetSuite, MySQL
BI Tools	Alteryx, Azure Data Factory, Databricks, Tableau Desktop Version (8.x/9.x/10.x/2019)/Tableau Server/Tableau Online, Power BI, SSIS, MS Excel (Pivot Table, VLOOKUP)
Hadoop Ecosystem	Hadoop 2, HDFS, HIVE, Pig, Spark 2.0
Languages	SQL, T-SQL, Snowflake, PL/SQL, Shell Script, Java, Python (2.x/3.x), VBS
Operating Systems	Windows 10/8/7/XP
Cloud Services	Microsoft Azure Blob/Logic App/Data Factory, Databricks, Amazon Web Services (AWS) S3/RDS/Redshift/EC2/EMR
Machine Learning Algorithms	Linear Regression, Logistic Regression, Naïve Bayes, Decision Tree, K-nearest Neighbors (KNN), K-means Clustering, Support Vector Machine (SVM), Neural Network
Others	SharePoint, Microsoft Team, Notepad++, Redgate, Beyond Compare, LucidChart, FileZilla, Jira, SAS, SSPS, Google Shared Services, Slack, JMP

Professional Experience

Body Armor Sports Nutrition, LLC, Queens, NY

May. 2020-Now

Data Engineer

A sports drink brand under The Coca-Cola Company. Responsible for supporting and maintaining ETL workflows that ingest, transform, and publish data from diverse source systems into centralized data warehouses for downstream Tableau reporting and analytics.

- Design, build, and optimize scalable cloud data pipelines and enterprise data warehouse solutions to support business intelligence and analytics across the organization.
- Engineer and maintain **200+ automated ETL/ELT pipelines** using Azure Data Factory, Alteryx, and SQL, processing **600M+ rows daily** from diverse data sources into Snowflake and SQL Server data warehouses.
- Improve data pipeline performance by **up to 40%** through query optimization, indexing strategies,

partitioning, and incremental loading design.

- Design and implement a reusable pipeline framework to standardize data ingestion processes, improving development efficiency and ensuring consistent data quality across multiple data sources.
- Build robust data infrastructure to support the end-to-end data lifecycle including ingestion, transformation, validation, and publishing for enterprise reporting.
- Develop complex T-SQL solutions including stored procedures, CTEs, window functions, indexing strategies, and performance tuning to enhance reliability and efficiency.
- Partner with cross-functional stakeholders to translate business requirements into technical data solutions and scalable reporting architectures.
- Deliver and maintain Tableau dashboards supporting **200+ business users**, enabling data-driven decision making across operations, finance, marketing, and HR teams.
- Lead troubleshooting and root-cause analysis for pipeline failures, data inconsistencies, and performance bottlenecks, ensuring high system reliability and minimal downtime.
- Migrate and deploy ETL workflows across cloud environments (SSMS/Alteryx/Azure), improving deployment efficiency and pipeline stability.
- Implement automated validation and monitoring processes to proactively detect data quality issues and improve pipeline resilience.

Environment: SQL Server Management Studio 20, T-SQL, Snowflake, Alteryx Designer/Scheduler, Azure (Azure Storage Explorer, Logic app, Azure Data Factory, Azure SQL Database Managed Instance, etc.), Tableau Desktop/Server/Online, NetSuite, Jira

J & M Imports

Aug. 2019-Dec. 2019

Business Analyst Consultant Intern

A regional used car purchasing and resale company, J & M Imports focuses on vehicle acquisition, inventory management, and local automotive sales operations

- Analyzed multi-branch sales data to identify trends, customer behavior patterns, and product performance insights for business decision support.
- Consolidated and integrated data from multiple sources into a centralized database to improve reporting efficiency and data accessibility.
- Performed data cleansing, transformation, and validation using Excel and Access to ensure data accuracy and consistency.
- Developed ad-hoc reports and visualizations to support management in strategic planning and operational decision-making.

Analysis for Elderly Care Environment on Alzheimer's Disease, University of Michigan

Jan. 2019-Jun. 2019

Capstone Project Team lead

A research initiative at University of Michigan focused on analyzing environmental factors in elderly care settings and their impact on individuals with Alzheimer's disease, supporting data-driven insights for improving care quality and living environments.

- Participated in all phases of project life cycle including data collection, data cleaning, validation, data mining, and reports creating.
- Designed a model to determine the factors which have higher possibility for Alzheimer's disease.
- Cleaned dataset which contains 59,025 records from National Alzheimer's Coordinating Center

(NACC) by using SQL&SAS and conducted feature selection by using Multiple Imputation in SPSS.

- Applied Regression Model, Boosted Trees, Bootstrap, and Neural Networks by using JMP (SAS based environment) and selected the best model, Nominal Logistic Regression Model, and visualized data in Tableau.
- Provided recommendations based on the regression model results to help people detect Alzheimer's disease in early phase.
- Utilized Control flow and Data flow components in SSIS, such as Pivot Transformation, Fuzzy Lookup, Derived Columns, Condition Split, Aggregate, Execute SQL Task, and Execute Package Task to build ETL pipelines.
- Created objects like Temp tables, Views, Stored Procedure, User-Defined Functions, DML Triggers, Clustered and Non-Clustered Indexes, Partition, and CTE using T-SQL in SQL Server 2012/2017 to store data and maintain data integrity.
- Optimized the performance of queries with modification in T-SQL to set constraints, remove redundant data, establish joins among tables.
- Conducted data blending, data preparation using SQL for Tableau consumption and publishing dashboard to Tableau Server.
- Built Stored Procedure for ETL in SQL and scheduled data feeding the deployed ETL packages using message queue monitoring to capture data change.
- Developed business intelligence Dashboards to produce different summary results based on requirements and role members using Bar charts, Line charts, Boxplot, and table contents.
- Applied EDA (Exploratory Data Analysis) to find data patterns, distributions, and correlations among different features of the customer data using Python Jupyter Notebook.

COSCO SHIPPING Financial Holdings Co., Ltd.

Sep 2016 – Sep 2017

Financial Analyst

A large state-owned international maritime transportation company under China COSCO Shipping Corporation, providing global ocean shipping, logistics, and supply chain services across major international trade routes.

- Performed full-cycle accounting and financial data management using UFIDA (SAP-based ERP system), ensuring data accuracy and reporting integrity.
- Analyzed financial statements to evaluate company performance and identify key financial risks and trends.
- Developed budget forecasts based on historical financial data and performance indicators to support annual planning.
- Utilized advanced Excel analytics (Pivot Tables, VLOOKUP, Index/Match) to conduct data analysis and generate actionable insights.
- Conducted deep-dive financial analysis to identify root causes of performance issues and recommended improvement strategies.
- Designed financial monitoring dashboards and presented analytical findings to senior management to support data-driven decision-making.

Education

MS in Business Analytics, University of Michigan, MI